Variant 11: Structural Separation of Formal Occurrence and Real Determination in a Binary Boolean Function

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May 22, 2026

Abstract

This paper considers the binary Boolean function

$$f_{11}(p, q) = \neg q$$

within the minimal setting of Boolean functions of two variables. The presentation of the function as $f_{11}(p, q)$ fixes its arity and the positions of its arguments, but it does not determine how the value of the function depends on the individual coordinates. The case examined here shows that these two descriptions need not coincide even in the binary setting.

The analysis provides a local description of this configuration, identifies the conditions under which it arises, and presents its algebraic form and local geometric representation within the adopted presentation.

Keywords: Boolean functions; binary Boolean functions; essential variables; fictitious variables; formal occurrence; real determination; essential arity; local factorization; Variant 11; projection factorization.
2020 MSC. Primary 06E30; Secondary 03B05.

1 Introduction

This paper studies a single binary Boolean function,

$$f_{11}(p, q) = \neg q.$$

As a Boolean function, Variant 11 is entirely classical. The purpose of the paper is not to introduce a new Boolean operation, but to isolate a structural distinction that becomes visible in its binary presentation.

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The guiding question is which of the variables occurring in a presentation actually participate in determining the value of a function. A function may be written with several formal variables, while its value may depend on only some of them. The paper examines this difference in the smallest setting in which it can be studied concretely: the finite Boolean setting of binary functions.

The distinction developed below is the distinction between *formal occurrence* and *real determination*. Formal occurrence concerns the variables present in the adopted presentation of a function. Real determination concerns the variables whose variation can affect the value of the function. Variant 11 will be studied as a binary Boolean function in which this distinction becomes visible.

The analysis is deliberately local. It remains internal to the finite Boolean environment fixed below and concerns only the structural relation between the binary presentation of Variant 11 and the variables that determine its value. No external interpretation is attached to the object under study.

We begin by fixing the finite Boolean environment in which Variant 11 is defined and located. We then introduce Variant 11 as the eleventh element of the fixed list of binary Boolean functions. The subsequent sections develop the real/fictitious split, distinguish formal occurrence from real determination, locate Variant 11 within its immediate binary environment, and record its local factorization and geometric reading.

2 Preliminaries

This section fixes the minimal Boolean environment used throughout the paper. The word *minimal* refers here only to the technical setting: the Boolean values, the binary input space, the order of inputs, and the complete list of binary Boolean functions.

No additional structure is assumed at this stage.

2.1 The Boolean Square

We work over the Boolean set

$$\{0,1\}.$$

The binary input space is

$$\{0,1\}^2.$$

Its elements are written as ordered pairs

$$(p,q),$$

where

$$p,q \in \{0,1\}.$$

Thus

$$\{0,1\}^2 = \{(0,0),(0,1),(1,0),(1,1)\}.$$

We shall refer to this set as the Boolean square. The coordinates of the Boolean square are denoted by p and q.

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Throughout the paper, the input order is fixed as

$$(0,0),(0,1),(1,0),(1,1).$$

This convention will be used in all truth tables and truth vectors.

2.2 Binary Boolean Functions

A binary Boolean function is a map

$$f : \{0,1\}^2 \to \{0,1\}.$$

We denote the set of all such functions by

$$\mathcal{F}_2 = \{f : \{0,1\}^2 \to \{0,1\}\}.$$

Since the domain $\{0,1\}^2$ has four elements and each element may be assigned either 0 or 1, the number of binary Boolean functions is

$$|\mathcal{F}_2| = 2^4 = 16.$$

Thus the complete finite class $\mathcal{F}_2$ consists of sixteen functions.

2.3 Truth Vectors and Variants

Given

$$f \in \mathcal{F}_2,$$

the truth vector of f, relative to the fixed input order

$$(0,0),(0,1),(1,0),(1,1),$$

is

$$[f(0,0), f(0,1), f(1,0), f(1,1)].$$

Equivalently, if

$$[a,b,c,d] \in \{0,1\}^4,$$

then the entries mean

$$a = f(0,0), \qquad b = f(0,1), \qquad c = f(1,0), \qquad d = f(1,1).$$

In this fixed input order, each binary Boolean function is represented by a unique truth vector in $\{0,1\}^4$.

The term *Variant* refers to the position of a truth vector in the fixed list of the sixteen truth vectors

$$[0,0,0,0],[0,0,0,1],\ldots,[1,1,1,1],$$

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ordered lexicographically. Thus Variant 1 corresponds to

$$[0, 0, 0, 0],$$

and Variant 16 corresponds to

$$[1, 1, 1, 1].$$

Equivalently, the vector

$$[a, b, c, d]$$

occupies position

$$1 + 8a + 4b + 2c + d$$

in this list.

This numbering is only a naming convention for the sixteen elements of $\mathcal{F}_2$.

2.4 The Complete Binary List

For reference, Table 1 displays the complete list of binary Boolean functions in the above convention. The formulas in the last column are representative formulas for the corresponding truth vectors. No structural interpretation is attached to this list at this stage.

Variant	Truth vector	Representative formula
1	$[0, 0, 0, 0]$	0
2	$[0, 0, 0, 1]$	$p \wedge q$
3	$[0, 0, 1, 0]$	$p \wedge \neg q$
4	$[0, 0, 1, 1]$	p
5	$[0, 1, 0, 0]$	$\neg p \wedge q$
6	$[0, 1, 0, 1]$	q
7	$[0, 1, 1, 0]$	$p \oplus q$
8	$[0, 1, 1, 1]$	$p \vee q$
9	$[1, 0, 0, 0]$	$\neg p \wedge \neg q$
10	$[1, 0, 0, 1]$	$p \leftrightarrow q$
11	$[1, 0, 1, 0]$	$\neg q$
12	$[1, 0, 1, 1]$	$p \vee \neg q$
13	$[1, 1, 0, 0]$	$\neg p$
14	$[1, 1, 0, 1]$	$\neg p \vee q$
15	$[1, 1, 1, 0]$	$\neg(p \wedge q)$
16	$[1, 1, 1, 1]$	1

Table 1: The sixteen binary Boolean functions in the fixed input order $(0, 0), (0, 1), (1, 0), (1, 1)$.

Here $\oplus$ denotes exclusive disjunction, and $\leftrightarrow$ denotes Boolean equivalence.

The next section isolates the eleventh element of this fixed list.

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3 Variant 11

With the notation and input order fixed above, we now introduce the object studied in this paper.

3.1 Definition of Variant 11

Definition. Variant 11 is the Boolean function

$$f_{11} : \{0,1\}^2 \to \{0,1\}$$

defined by

$$\boxed{f_{11}(p,q) = \neg q.}$$

Thus the value of f_{11} is obtained by negating the second variable. The first variable p remains present in the binary presentation of the function, while the displayed defining expression depends only on q.

With respect to the input order

$$(0,0),(0,1),(1,0),(1,1),$$

the truth vector of Variant 11 is

$$[1,0,1,0].$$

Indeed,

$$f_{11}(0,0) = 1, \qquad f_{11}(0,1) = 0,$$

and

$$f_{11}(1,0) = 1, \qquad f_{11}(1,1) = 0.$$

3.2 Truth Table

The truth table of Variant 11 is shown in Table 2.

p	q	$f_{11}(p,q)$
0	0	1
0	1	0
1	0	1
1	1	0

Table 2: Truth table of Variant 11.

The table shows that, for a fixed value of q, changing p does not change the value of the function. For $q = 0$, both rows give the value 1; for $q = 1$, both rows give the value 0.

By contrast, changing q changes the output. When $q = 0$, the value is 1; when $q = 1$, the value is 0, independently of p.

At this stage we only record this as a feature visible in the truth table. The formal distinction between fictitious and real variables will be introduced in the next section.

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3.3 Logical Reduction

Variant 11 can also be written in disjunctive normal form. Since the value of f_{11} is 1 exactly at the input pairs

$$(0,0) \qquad \text{and} \qquad (1,0),$$

we obtain

$$f_{11}(p,q) = (\neg p \wedge \neg q) \vee (p \wedge \neg q).$$

Factoring out $\neg q$, we get

$$f_{11}(p,q) = \neg q \wedge (\neg p \vee p).$$

Since

$$\neg p \vee p = 1,$$

it follows that

$$f_{11}(p,q) = \neg q.$$

Thus the expanded binary expression reduces, as an expression for the value, to $\neg q$:

$$(\neg p \wedge \neg q) \vee (p \wedge \neg q) = \neg q.$$

The reduction exhibits the same feature already visible in the truth table: the function is presented with two formal variables, while its value is determined by the second one. The following sections will make this distinction precise.

4 Real and Fictitious Variables

The preceding section introduced Variant 11 as the Boolean function

$$f_{11}(p,q) = \neg q.$$

In order to describe its internal structure precisely, we now introduce the distinction between fictitious and real variables.

This distinction is always relative to a given function. A variable is not real or fictitious in isolation. It is real or fictitious only with respect to the way it participates in the determination of the value of a function.[1]

4.1 Fictitious Variables

Let

$$f : \{0,1\}^2 \to \{0,1\}$$

be a Boolean function of two variables, written as

$$f(p,q).$$

[1]The dependence criterion used below is classical. In standard terminology, a variable is essential if changing it, while the remaining variables are fixed, can change the value of the function; otherwise it is inessential. See, for example, [1, 2]. The terms *real* and *fictitious* are used in the present paper as local determinational terminology for the two sides of this classical distinction.

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Definition. The variable p is called *fictitious relative to* f if changing p, while keeping q fixed, never changes the value of the function. Formally,

$$p \in \mathrm{Fict}(f) \quad \Longleftrightarrow \quad \forall q \in \{0,1\},\ f(0,q) = f(1,q).$$

Similarly, the variable q is called *fictitious relative to* f if changing q, while keeping p fixed, never changes the value of the function. Formally,

$$q \in \mathrm{Fict}(f) \quad \Longleftrightarrow \quad \forall p \in \{0,1\},\ f(p,0) = f(p,1).$$

Thus a fictitious variable may occur in the notation of a function without participating in the determination of its value.

Equivalently, a variable is fictitious relative to f when the truth table of f is constant along the direction corresponding to that variable.

4.2 Real Variables

A variable is real relative to a function if it is not fictitious.

Definition. The variable p is called *real relative to* f if changing p, while keeping q fixed, can change the value of the function. Formally,

$$p \in \mathrm{Real}(f) \quad \Longleftrightarrow \quad \exists q \in \{0,1\},\ f(0,q) \neq f(1,q).$$

Similarly, the variable q is called *real relative to* f if changing q, while keeping p fixed, can change the value of the function. Formally,

$$q \in \mathrm{Real}(f) \quad \Longleftrightarrow \quad \exists p \in \{0,1\},\ f(p,0) \neq f(p,1).$$

In the present two-variable setting, this is equivalently expressed as

$$\mathrm{Real}(f) = \{p, q\} \setminus \mathrm{Fict}(f).$$

A real variable is therefore a variable that participates in determining the value of f. A change in that variable can be reflected in the output of the function.

4.3 Variables, Not Values

The distinction between real and fictitious is a distinction between variables relative to a function, not a distinction between values of the function.

Thus, for a Boolean function

$$f : \{0,1\}^2 \to \{0,1\},$$

the values 0 and 1 are not called real or fictitious. They are simply the possible values of the function.

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What may be real or fictitious is a variable, and only relative to a given function. In the present setting, this means that p or q may be real or fictitious relative to f, depending on whether changing that variable can change the value of f.

In short, the real/fictitious distinction applies to variables relative to a function, not to function values.

This local and functional meaning of the terms *real* and *fictitious* will be used in the next section to determine the exact real/fictitious split of Variant 11.

5 The Real/Fictitious Split of Variant 11

We now apply the definitions of Section 4 to Variant 11.

Recall that Variant 11 is the Boolean function

$$f_{11} : \{0,1\}^2 \to \{0,1\}$$

defined by

$$f_{11}(p,q) = \neg q.$$

The purpose of this section is to determine exactly which variable is fictitious and which variable is real relative to f_{11}.

5.1 Fictitiousness of p

We first show that p is fictitious relative to f_{11}.

By definition, p is fictitious relative to f_{11} if

$$\forall q \in \{0,1\},\ f_{11}(0,q) = f_{11}(1,q).$$

For every $q \in \{0,1\}$, we have

$$f_{11}(0,q) = \neg q$$

and

$$f_{11}(1,q) = \neg q.$$

Therefore,

$$f_{11}(0,q) = f_{11}(1,q)$$

for every $q \in \{0,1\}$.

It follows that changing the value of p, while keeping q fixed, never changes the value of f_{11}. Hence

$$p \in \mathrm{Fict}(f_{11}).$$

Thus p occurs in the binary presentation of f_{11}, but it is fictitious relative to the value of the function.

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5.2 Realness of q

We now show that q is real relative to f_{11}.

By definition, q is real relative to f_{11} if changing q, while keeping p fixed, can change the value of the function. Equivalently,

$$\exists p \in \{0,1\},\ f_{11}(p,0) \neq f_{11}(p,1).$$

In fact, for every $p \in \{0,1\}$, we have

$$f_{11}(p,0) = 1$$

and

$$f_{11}(p,1) = 0.$$

Therefore,

$$f_{11}(p,0) \neq f_{11}(p,1)$$

for every $p \in \{0,1\}$.

It follows that changing the value of q, while keeping p fixed, changes the value of f_{11}. Hence

$$q \in \mathrm{Real}(f_{11}).$$

Thus q is the variable that participates in the determination of the value of f_{11}.

5.3 The Split

The preceding two subsections give the exact real/fictitious split of Variant 11.

We now summarize the results of Subsections 5.1 and 5.2.

Theorem 5.1 (The real/fictitious split of Variant 11)**.** *For Variant 11,*

$$f_{11}(p,q) = \neg q,$$

one has

$$\mathrm{Fict}(f_{11}) = \{p\}$$

and

$$\mathrm{Real}(f_{11}) = \{q\}.$$

Proof. By Subsection 5.1, changing p, while keeping q fixed, never changes the value of f_{11}. Hence

$$p \in \mathrm{Fict}(f_{11}).$$

By Subsection 5.2, changing q, while keeping p fixed, changes the value of f_{11}. Hence

$$q \in \mathrm{Real}(f_{11}).$$

Since the only variables in the present binary presentation are p and q, it follows that

$$\mathrm{Fict}(f_{11}) = \{p\}$$

and

$$\mathrm{Real}(f_{11}) = \{q\}.$$

□

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Equivalently, the full split may be written as

$$\mathrm{Fict}(f_{11}) = \{p\}, \qquad \mathrm{Real}(f_{11}) = \{q\}.$$

This is the central structural fact about Variant 11: the function is written with two variables, but only one of them is real relative to the function.

6 Syntactically Binary, Determinationally Unary

The real/fictitious split proved in Section 5 has an immediate structural consequence.

Variant 11 is written as a Boolean function of two variables:

$$f_{11}(p, q) = \neg q.$$

Thus its presentation is syntactically binary: the function is presented with two formal variables, p and q.

However, Section 5 showed that only one of these variables is real relative to f_{11}:

$$\mathrm{Real}(f_{11}) = \{q\}.$$

The other variable is fictitious:

$$\mathrm{Fict}(f_{11}) = \{p\}.$$

Thus the syntactic arity of the presentation and the number of real variables do not coincide.

6.1 Arity in the Adopted Binary Presentation

The adopted binary presentation of Variant 11 contains two variables:

$$p \qquad \text{and} \qquad q.$$

In this adopted presentation, Variant 11 has arity 2.

We write this as

$$\mathrm{arity}(f_{11}) = 2.$$

This statement refers only to the binary presentation of the function:

$$f_{11}(p, q).$$

It does not yet say which of the two variables participates in the determination of the value.

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6.2 Determinational Unarity

The real/fictitious split gives the determinational content of Variant 11:

$$\mathrm{Real}(f_{11}) = \{q\}.$$

For notational convenience, we write

$$\mathrm{GD}(f) \coloneqq |\,\mathrm{Real}(f)|.$$

This number records only how many variables are real relative to f. No additional structure is attached to this notation.

For Variant 11, we therefore have

$$\mathrm{GD}(f_{11}) = |\,\mathrm{Real}(f_{11})| = 1.$$

Thus, although Variant 11 is presented with two variables, its value is determined by exactly one of them.

6.3 The Central Formula

Combining the two preceding observations, we obtain:

$$\mathrm{arity}(f_{11}) = 2, \qquad \mathrm{GD}(f_{11}) = 1.$$

Equivalently,

$$\mathrm{arity}(f_{11}) = 2, \qquad |\,\mathrm{Real}(f_{11})| = 1.$$

This is the precise sense in which Variant 11 is syntactically binary but determinationally unary:

Variant 11 is syntactically binary but determinationally unary.

The function is syntactically binary because it is presented as $f_{11}(p, q)$. It is determinationally unary because only q participates in the determination of its value.

This formulation will serve as the bridge to the next section, where the distinction between formal occurrence and real determination will be made explicit.

7 Formal Occurrence and the Determinational Viewpoint

The preceding section showed that Variant 11 is syntactically binary but determinationally unary:

$$\mathrm{arity}(f_{11}) = 2, \qquad |\mathrm{Real}(f_{11})| = 1.$$

This means that the binary presentation of the function contains two formal variables, while only one of them participates in the determination of its value.

This section makes explicit the distinction already implicit in this fact: the distinction between formal occurrence and real determination.

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7.1 Formal Occurrence

Variant 11 is presented as

$$f_{11}(p, q) = \neg q.$$

In this presentation, both p and q occur formally.

We write, locally for Variant 11,

$$\mathrm{Occ}(f_{11}) = \{p, q\}.$$

Here $\mathrm{Occ}(f_{11})$ denotes the variables occurring in the adopted binary presentation of the function.

This notation is used only to record the formal occurrence of variables in the presentation

$$f_{11}(p, q).$$

It is not introduced here as the beginning of a general theory of occurrence.

Thus,

$$p \in \mathrm{Occ}(f_{11})$$

and

$$q \in \mathrm{Occ}(f_{11}).$$

Formal occurrence records which variables appear in the notation of the function. It does not, by itself, record which of those variables participate in the determination of the value.

7.2 Real Determination

By Theorem 5.1, the real/fictitious split of Variant 11 is

$$\mathrm{Fict}(f_{11}) = \{p\}, \qquad \mathrm{Real}(f_{11}) = \{q\}.$$

Thus q is real relative to f_{11}, whereas p is fictitious relative to f_{11}.

Equivalently,

$$q \in \mathrm{Occ}(f_{11}) \quad \text{and} \quad q \in \mathrm{Real}(f_{11}),$$

whereas

$$p \in \mathrm{Occ}(f_{11}) \quad \text{but} \quad p \notin \mathrm{Real}(f_{11}).$$

Proposition 7.1 (Separation of formal occurrence and real determination)**.** *For Variant 11,*

$$\boxed{\mathrm{Real}(f_{11}) \subsetneq \mathrm{Occ}(f_{11})}$$

More explicitly,

$$\mathrm{Occ}(f_{11}) = \{p, q\}, \qquad \mathrm{Real}(f_{11}) = \{q\}.$$

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Proof. By the adopted binary presentation,

$$\mathrm{Occ}(f_{11}) = \{p, q\}.$$

By Theorem 5.1,

$$\mathrm{Real}(f_{11}) = \{q\}.$$

Since

$$\{q\} \subsetneq \{p, q\},$$

the result follows. □

The proposition shows that the variables occurring formally in the presentation of Variant 11 do not coincide with the variables that determine its value. This is the precise point at which the distinction between notation and determination becomes visible.

7.3 The Determinational Viewpoint

We now contrast the extensional and determinational viewpoints.

A Boolean function may be studied extensionally, through the values it assigns to input tuples. In this sense, Variant 11 is completely specified by its truth vector

$$[1, 0, 1, 0].$$

One may, however, ask a different question: which variables, among those occurring in the presentation of the function, participate in determining its value?

For Variant 11, this question does not change the function. It changes what is isolated in the function.

The extensional description gives the values:

$$[1, 0, 1, 0].$$

The determinational description gives the split:

$$\mathrm{Occ}(f_{11}) = \{p, q\}, \qquad \mathrm{Real}(f_{11}) = \{q\}, \qquad \mathrm{Fict}(f_{11}) = \{p\}.$$

Thus Variant 11 is not introduced here as a new Boolean function. It is isolated as a binary presentation in which formal occurrence and real determination separate.

The Boolean calculus itself remains unchanged. The function

$$f_{11}(p, q) = \neg q$$

is a classical Boolean function. The determinational viewpoint exposes, however, that the formal notation of a function and its determinational structure need not coincide.

In this precise sense, Variant 11 makes a crack not in Boolean calculus itself, but in the naive identification of the notation of a function with its real determinational structure.

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8 The Visibility of the Separation

Proposition 7.1 established the strict separation

$$\mathrm{Real}(f_{11}) \subsetneq \mathrm{Occ}(f_{11}).$$

More explicitly,

$$\mathrm{Occ}(f_{11}) = \{p, q\}, \qquad \mathrm{Real}(f_{11}) = \{q\}.$$

The variable p therefore occupies a precise but limited position in the structure of Variant 11:

$$p \in \mathrm{Occ}(f_{11}), \qquad p \notin \mathrm{Real}(f_{11}).$$

It occurs formally in the adopted binary presentation, but it does not participate in determining the value of the function.

To avoid ambiguity, the term *necessary* is used here in two distinct and explicitly restricted senses.

First, p is not necessary for determining the value of f_{11}. For every $p, q \in \{0, 1\}$,

$$f_{11}(p, q) = \neg q.$$

The value of the function is therefore completely determined by q.

Second, retaining p as a formally occurring variable is necessary within the adopted binary presentation if the strict separation between formal occurrence and real determination is to remain visible. It is precisely the relation

$$p \in \mathrm{Occ}(f_{11}) \setminus \mathrm{Real}(f_{11})$$

that exhibits a variable occurring formally without participating in the determination of the value.

To make this distinction explicit, consider the unary Boolean function

$$g : \{0, 1\} \to \{0, 1\}$$

defined by

$$g(q) = \neg q.$$

The functions f_{11} and g are not identical: their domains are different. The function g records only the unary q-dependent output rule of Variant 11.

In the unary function g, the only variable occurring in the presentation is q, and that same variable determines the value. There is therefore no formally occurring variable that fails to participate in the determination of the output.

Passing from

$$f_{11}(p, q) = \neg q$$

to

$$g(q) = \neg q$$

thus replaces the binary function by a different unary function while preserving the q-dependent output rule. What is no longer present is the binary structural context in which formal occurrence and real determination can diverge.

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Remark 8.1 (Visibility of the separation). The variable p remains fully fictitious relative to f_{11}. It is not necessary for determining the value of the function, and no hidden determinational role or second kind of realness is assigned to it.

Its formal presence is necessary only in the restricted sense that, within the adopted binary presentation, it makes possible the explicit relation

$$p \in \mathrm{Occ}(f_{11}) \setminus \mathrm{Real}(f_{11}).$$

This necessity concerns the role of a formally occurring fictitious variable, not the particular symbol p.

Thus fictitiousness relative to the value does not imply irrelevance to the visibility of the structure. The separation belongs to Variant 11 considered as a binary function; its visibility is lost when one passes to the associated unary function g.

9 Conditional Uniqueness and Minimality

The preceding sections established the real/fictitious split of Variant 11, the distinction between formal occurrence and real determination, and the visibility of this separation in the adopted binary presentation.

We now locate Variant 11 within its immediate Boolean environment and determine the precise senses in which it is unique and minimal. The classification serves to locate the object, not to reduce the distinction isolated above.

The two questions are distinct. Conditional uniqueness concerns the selection of Variant 11 from a fixed determinational role configuration. Minimality concerns the least arity at which a real variable and a fictitious variable can coexist in a nonconstant Boolean function.

9.1 Determinational Role Configurations in the Binary Setting

In this section, the *determinational role configuration* of a binary Boolean function f refers to the ordered pair

$$\Big(\mathrm{Real}(f), \mathrm{Fict}(f)\Big).$$

The following proposition classifies the binary Boolean functions having exactly one real variable and one fictitious variable.

Proposition 9.1. *Let*

$$f \in \mathcal{F}_2.$$

Then f has exactly one real variable and exactly one fictitious variable if and only if it is one of the four functions

$$f(p,q) = p, \qquad f(p,q) = \neg p, \qquad f(p,q) = q, \qquad f(p,q) = \neg q.$$

These four functions realize exactly two determinational role configurations. The configuration

$$\mathrm{Real}(f) = \{p\}, \qquad \mathrm{Fict}(f) = \{q\}$$

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is realized by

$$f(p,q) = p \qquad \textit{and} \qquad f(p,q) = \neg p,$$

whereas the configuration

$$\mathrm{Real}(f) = \{q\}, \qquad \mathrm{Fict}(f) = \{p\}$$

is realized by

$$f(p,q) = q \qquad \textit{and} \qquad f(p,q) = \neg q.$$

Proof. Suppose first that

$$\mathrm{Real}(f) = \{p\}, \qquad \mathrm{Fict}(f) = \{q\}.$$

Since q is fictitious relative to f,

$$f(p,0) = f(p,1)$$

for every $p \in \{0,1\}$. Hence the value of f depends only on p.

Since p is real relative to f, this dependence is nonconstant. The only nonconstant Boolean functions of the single variable p are

$$p \qquad \text{and} \qquad \neg p.$$

Therefore,

$$f(p,q) = p \qquad \text{or} \qquad f(p,q) = \neg p.$$

Similarly, suppose that

$$\mathrm{Real}(f) = \{q\}, \qquad \mathrm{Fict}(f) = \{p\}.$$

Since p is fictitious relative to f,

$$f(0,q) = f(1,q)$$

for every $q \in \{0,1\}$. Hence the value of f depends only on q.

Since q is real relative to f, this dependence is nonconstant. The only nonconstant Boolean functions of the single variable q are

$$q \qquad \text{and} \qquad \neg q.$$

Therefore,

$$f(p,q) = q \qquad \text{or} \qquad f(p,q) = \neg q.$$

Conversely, each of the four functions listed above has exactly one real variable and exactly one fictitious variable, with the determinational role configurations stated above. □

Proposition 9.1 identifies four Boolean functions but only two determinational role configurations. The functions

$$f(p,q) = p \qquad \text{and} \qquad f(p,q) = \neg p$$

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are two Boolean realizations of the same role configuration, whereas the functions

$$f(p,q) = q \qquad \text{and} \qquad f(p,q) = \neg q$$

are two Boolean realizations of the other.

The purpose of this classification is to locate Variant 11 within its immediate binary environment. It distinguishes the level of Boolean realization from the level of determinational role configuration.

For the role configuration established for Variant 11,

$$\mathrm{Fict}(f) = \{p\}, \qquad \mathrm{Real}(f) = \{q\},$$

Proposition 9.1 gives

$$f(p,q) = q \qquad \text{or} \qquad f(p,q) = \neg q.$$

Thus the real/fictitious split alone does not yet select Variant 11. It determines a two-element class of Boolean realizations.

9.2 Conditional Selection of Variant 11

Within the fixed determinational role configuration

$$\mathrm{Fict}(f) = \{p\}, \qquad \mathrm{Real}(f) = \{q\},$$

Variant 11 can be selected by one additional valuational condition.

Corollary 9.2 (Conditional uniqueness of Variant 11)**.** *Let*

$$f \in \mathcal{F}_2.$$

Suppose that

$$\mathrm{Fict}(f) = \{p\}, \qquad \mathrm{Real}(f) = \{q\},$$

and

$$f(0,0) = 1.$$

Then

$$f = f_{11},$$

where

$$f_{11}(p,q) = \neg q.$$

Proof. By Proposition 9.1, the assumed real/fictitious split implies that

$$f(p,q) = q \qquad \text{or} \qquad f(p,q) = \neg q.$$

For the first realization,

$$f(p,q) = q,$$

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one has

$$f(0,0) = 0.$$

For the second realization,

$$f(p,q) = \neg q,$$

one has

$$f(0,0) = 1.$$

The condition

$$f(0,0) = 1$$

therefore selects

$$f(p,q) = \neg q.$$

Hence

$$f = f_{11}.$$

□

Remark 9.3 (The role of the valuational condition). The condition

$$f(0,0) = 1$$

is used solely as a convenient selector between the two Boolean realizations of the fixed determinational role configuration

$$\mathrm{Fict}(f) = \{p\}, \qquad \mathrm{Real}(f) = \{q\}.$$

No semantic or ontological meaning beyond this technical selecting role is attached to the condition. The uniqueness established in Corollary 9.2 is therefore explicitly conditional.

The localization proceeds through three distinct levels:

$$\{p, \neg p, q, \neg q\},$$

the four binary Boolean functions having exactly one real and one fictitious variable;

$$\{q, \neg q\},$$

the two realizations of the configuration in which p is fictitious and q is real; and finally

$$\{\neg q\},$$

the single realization selected by the technical condition

$$f(0,0) = 1.$$

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9.3 Minimal Arity for the Coexistence of Real and Fictitious Variables

The minimality considered here does not concern the length of a formula or the mere inequality between formal occurrence and real determination. It concerns the least arity at which a nonconstant Boolean function can contain both a real variable and a fictitious variable.

Theorem 9.4 (Minimality of the binary setting)**.** *Among Boolean functions of positive arity, arity* 2 *is the least arity for which a nonconstant Boolean function can have both a real variable and a fictitious variable.*

Proof. Consider first a unary Boolean function

$$h : \{0,1\} \to \{0,1\}$$

with the single variable x.

For the adopted unary presentation $h(x)$, the single formal variable x is fictitious relative to h exactly when changing its value does not change the value of the function, and real relative to h exactly when it does. Thus, in this unary setting,

$$x \in \mathrm{Fict}(h) \quad \Longleftrightarrow \quad h(0) = h(1),$$

and

$$x \in \mathrm{Real}(h) \quad \Longleftrightarrow \quad h(0) \neq h(1).$$

There are two possible cases.

If h is constant, then changing x never changes the value of the function. Hence

$$\mathrm{Real}(h) = \varnothing, \qquad \mathrm{Fict}(h) = \{x\}.$$

If h is nonconstant, then changing x changes the value of the function. Hence

$$\mathrm{Real}(h) = \{x\}, \qquad \mathrm{Fict}(h) = \varnothing.$$

Thus no unary Boolean function can have both a real variable and a fictitious variable.

By contrast, Variant 11 is a nonconstant binary Boolean function and satisfies

$$\mathrm{Real}(f_{11}) = \{q\}, \qquad \mathrm{Fict}(f_{11}) = \{p\}.$$

Therefore arity 2 is the least positive arity at which a nonconstant Boolean function can contain both kinds of variables. □

Theorem 9.4 establishes the precise sense in which Variant 11 occupies a minimal setting.

Variant 11 is one of the four binary Boolean functions of minimal arity having exactly one real and one fictitious variable. It is one of the two Boolean realizations of the determinational role configuration in which p is fictitious and q is real. Within that fixed configuration, it is uniquely selected by the technical condition

$$f(0,0) = 1.$$

Variant 11 is therefore neither absolutely unique nor uniquely minimal. It is conditionally unique within a fixed determinational role configuration and situated at the minimal arity supporting the coexistence of real and fictitious variables.

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10 Local Factorization and Local Geometry

The preceding sections established that Variant 11 is a binary Boolean function with exactly one real variable and one fictitious variable:

$$\mathrm{Real}(f_{11}) = \{q\}, \qquad \mathrm{Fict}(f_{11}) = \{p\}.$$

This section records the same structure in two local forms. First, the dependence of f_{11} on its real coordinate is expressed through a coordinate factorization. Second, the real/fictitious split is read geometrically on the Boolean square $\{0,1\}^2$.

The constructions in this section are purely local and serve only to display the determinational structure of Variant 11 in algebraic and geometric form. Neither construction changes the status of Variant 11 as a binary Boolean function.

10.1 Projection onto the Real Coordinate

Consider the coordinate projection

$$\pi_q : \{0,1\}^2 \to \{0,1\}$$

defined by

$$\pi_q(p,q) = q.$$

The projection π_q retains the coordinate q and omits the coordinate p from the projected input. Since

$$\mathrm{Real}(f_{11}) = \{q\},$$

it preserves precisely the variable that participates in determining the value of Variant 11.

We also consider the unary Boolean function

$$g : \{0,1\} \to \{0,1\}$$

defined by

$$g(q) = \neg q.$$

The functions f_{11} and g are not identical. Their domains are different:

$$f_{11} : \{0,1\}^2 \to \{0,1\}, \qquad g : \{0,1\} \to \{0,1\}.$$

The function g records the unary q-dependent output rule associated with Variant 11.

10.2 Local Factorization

Proposition 10.1 (Local factorization of Variant 11)**.** *Let*

$$\pi_q : \{0,1\}^2 \to \{0,1\}, \qquad \pi_q(p,q) = q,$$

and let

$$g : \{0,1\} \to \{0,1\}, \qquad g(q) = \neg q.$$

Then

$$f_{11} = g \circ \pi_q.$$

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Proof. For every

$$(p, q) \in \{0, 1\}^2,$$

one has

$$(g \circ \pi_q)(p, q) = g\Big(\pi_q(p, q)\Big) = g(q) = \neg q = f_{11}(p, q).$$

Therefore,

$$f_{11} = g \circ \pi_q.$$

□

The factorization may be written schematically as

$$(p, q) \xrightarrow{\pi_q} q \xrightarrow{g} \neg q.$$

It records that the value of Variant 11 depends only on the projected coordinate q. It does not identify the binary function f_{11} with the unary function g, nor does it replace the binary domain of Variant 11 by the unary domain of g.

The projection π_q has two fibers:

$$\pi_q^{-1}(0) = \{(0, 0), (1, 0)\},$$

and

$$\pi_q^{-1}(1) = \{(0, 1), (1, 1)\}.$$

Variant 11 is constant on each fiber:

$$f_{11}(0, 0) = f_{11}(1, 0) = 1,$$

and

$$f_{11}(0, 1) = f_{11}(1, 1) = 0.$$

Thus input pairs that differ only in the fictitious coordinate p are mapped by f_{11} to the same value.

10.3 Local Geometric Reading

The domain

$$\{0, 1\}^2$$

may be viewed as the Boolean square with vertices

$$(0, 0), \qquad (0, 1), \qquad (1, 0), \qquad (1, 1).$$

Its two coordinate directions correspond to the variables p and q.

For every $q \in \{0, 1\}$,

$$f_{11}(0, q) = f_{11}(1, q).$$

Hence Variant 11 is constant along each edge of the Boolean square parallel to the p-coordinate direction.

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By contrast, for every $p \in \{0,1\}$,

$$f_{11}(p,0) \neq f_{11}(p,1).$$

More precisely,

$$f_{11}(p,0) = 1, \qquad f_{11}(p,1) = 0.$$

Hence Variant 11 is nonconstant along each edge parallel to the q-coordinate direction.

Remark 10.2 (Local geometric reading). On the Boolean square, the p-coordinate direction is the direction of constancy of Variant 11, whereas the q-coordinate direction is the direction of determination.

This gives a local geometric reading of the formal split

$$\mathrm{Fict}(f_{11}) = \{p\}, \qquad \mathrm{Real}(f_{11}) = \{q\}.$$

The geometric reading illustrates the already established real/fictitious structure; it does not replace its formal proof.

The local factorization and the local geometry therefore express the same determinational structure in two different forms:

$$f_{11} = g \circ \pi_q$$

records it algebraically, while constancy in the p-direction and variation in the q-direction record it geometrically.

Thus Section 10 closes the local algebraic and geometric display of Variant 11 without opening a separate theory of factorization or geometry.

11 Conclusion

This paper has isolated the structural content of the binary Boolean function

$$f_{11}(p,q) = \neg q.$$

As a Boolean function, Variant 11 remains entirely classical. Its role in the present analysis does not consist in introducing a new Boolean operation, but in making visible a distinction between formal occurrence and real determination.

The central fact established above is the strict separation

$$\mathrm{Occ}(f_{11}) = \{p,q\}, \qquad \mathrm{Real}(f_{11}) = \{q\}.$$

Thus

$$\mathrm{Real}(f_{11}) \subsetneq \mathrm{Occ}(f_{11}).$$

The variable p occurs in the adopted binary presentation, but it does not participate in determining the value of the function. The variable q is the unique real variable of f_{11}.

This is the precise sense in which Variant 11 is syntactically binary but determinationally unary. Its binary presentation contains two formal variables, while its value is determined by

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only one of them. The fictitious variable p is not necessary for the value of the function, but its formal occurrence is necessary for the visibility of the separation between occurrence and determination.

The classification of the immediate binary environment locates Variant 11 without reducing its significance. Variant 11 is not absolutely unique among binary Boolean functions with one real and one fictitious variable. Rather, it is conditionally selected within the fixed determinational role configuration

$$\mathrm{Fict}(f) = \{p\}, \qquad \mathrm{Real}(f) = \{q\},$$

and it lies at the minimal arity at which a nonconstant Boolean function can contain both a real variable and a fictitious variable.

The same structure admits a local algebraic and geometric display. Variant 11 is a binary Boolean function admitting the local factorization

$$\boxed{\begin{aligned} f_{11} &= g \circ \pi_q, \\ \pi_q(p,q) &= q, \qquad g(q) = \neg q. \end{aligned}}$$

The adopted binary presentation is structurally necessary for the visibility of this determinational structure. Only in the presence of the formally occurring fictitious variable p does the split

$$\mathrm{Occ}(f_{11}) = \{p, q\}, \qquad \mathrm{Real}(f_{11}) = \{q\}$$

become visible. In this precise sense, Variant 11 is the minimal binary structure in which formal occurrence and real determination strictly separate.

Geometrically, the same split is read on the Boolean square as constancy in the p-coordinate direction and variation in the q-coordinate direction. This geometric reading does not replace the formal result; it only displays it locally.

The analysis therefore stops at a local and purely structural point. Boolean calculus itself remains unchanged. What Variant 11 makes visible is that the notation of a function and its real determinational structure need not coincide.

A A Note on Further Context

The analysis in this paper is fully local. All definitions and arguments concern binary Boolean functions and require no additional framework.

Variant 11 may also serve as a finite example of the distinction between formal presentation and real determination. In a broader setting, the result

$$\mathrm{Real}(f_{11}) \subsetneq \mathrm{Occ}(f_{11})$$

illustrates a separation between what is formally present and what is determinationally active.

This observation has no role in the argument above. It is recorded only to indicate a possible further use of Variant 11 in an independent development.

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